6. CONVOLUTION AND SAMPLING TECHNIQUES

Objective

This experiment is divided into four simulations:

1.Convolution of Continuous Time Signals

2. Convolution of Discrete Time Signals

3. Sampling Theorem Verification

4. Aliasing Effect Demonstration

Experiment 6.1

Convolution of Continuous Time Signals

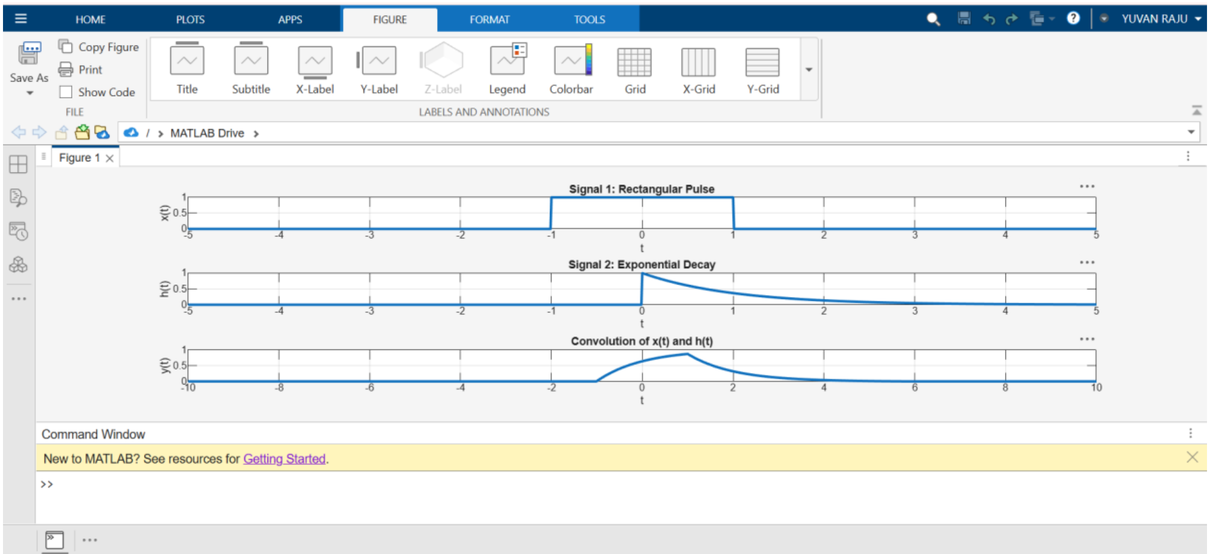
PROGRAM

```
// Test Case: Convolution of Continuous-Time Signals

// Define time range and sampling interval
t_min = -5; // Start time
t_max = 5; // End time
dt = 0.01; // Time step
t = t_min:dt:t_max; // Time vector
// Define Signal 1: Rectangular pulse of width 2 centered at t = 0
x_t = (t >= -1) .* (t <= 1);
// Define Signal 2: Exponential decay function
h_t = exp(-t) .* (t >= 0); // Exponential decay, causal
// Perform the convolution
y_t = conv(x_t, h_t) * dt; // Multiply by dt to scale correctly
// Adjust the time vector for the convolution result
t_conv = (2 * t_min):dt:(2 * t_max);
// Plot Signal 1
clf; // Clear the current figure
subplot(3,1,1);
plot(t, x_t, 'b');
xlabel('t');
```

```
ylabel('x(t)');
title('Signal 1: Rectangular Pulse');
// Plot Signal 2
subplot(3,1,2);
plot(t, h_t, 'g');
xlabel('t');
ylabel('h(t)');
title('Signal 2: Exponential Decay');
// Plot the Convolution Result
subplot(3,1,3);
plot(t_conv, y_t, 'r');
xlabel('t');
ylabel('y(t)');
title('Convolution of x(t) and h(t)');
```

OUTPUT



Experiment 6.2

Convolution of Discrete Time Signals

PROGRAM

```
% Convolution of Discrete-Time Signals
clc;
clear;
```

```
close all;

% Define the discrete-time signals
x_n = [1 2 3];       % Signal x[n]
h_n = [1 -1];        % Signal h[n]

% Compute the convolution
y_n = conv(x_n, h_n);

% Define the time index for the output signal
n_y = 0:(length(y_n)-1);

% Display the convolution result
disp('Convolution Output y[n]:');
disp(y_n);

% Plot the input signals and their convolution
figure;

% Plot x[n]
subplot(3,1,1);
stem(0:length(x_n)-1, x_n, 'filled');
grid on;
xlabel('n');
ylabel('x[n]');
title('Signal x[n]');

% Plot h[n]
subplot(3,1,2);
stem(0:length(h_n)-1, h_n, 'filled');
grid on;
```

```
xlabel('n');
ylabel('h[n]');
title('Signal h[n]');

% Plot y[n] (convolution result)
subplot(3,1,3);
stem(n_y, y_n, 'filled');
grid on;
xlabel('n');
ylabel('y[n]');
title('Convolution of x[n] and h[n]');
```

OUTPUT